LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Synchro torque pressure-indicating system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Synchro torque pressure-indicating system.
5. Specific Objectives	15.1.8.5 Trainee will be able to explain Synchro torque pressure indicating system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system schematic diagrams and verification of torque pressure indication principles against IB data; Safety: Ensure no live electrical connections are simulated; Supervision: Instructor monitors diagram interpretation accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a synchro torque system?
(b) How does pressure indication differ from direct pressure measurement?
(c) Why is torque required in pressure-indicating synchro systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• System overview and components • Synchro torque generation principles • Pressure sensing mechanism integration • Signal transmission and indication • System calibration and zeroing	L & Ds	PPT, IB	15 min
(b)	• Operational sequence analysis • Fault detection and isolation • Maintenance and troubleshooting procedures • Impact of torque failure on indication • Safety considerations during operation	L & Ds	PPT, PDF	20 min
(c)	• Real-world application scenarios • Comparison with alternative systems • Performance limitations and constraints • Future developments in torque systems • Regulatory compliance and standards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of torque in the pressure-indicating synchro system.
(b) Describe the steps involved in troubleshooting a torque failure.
(c) How does the system ensure accurate pressure indication under varying loads?

12. Home Assignment and Follow Up	Prepare a brief report comparing the synchro torque pressure-indicating system with the engine vibration monitoring system covered in LP 24, highlighting key operational differences.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine vibration monitoring and indicating system.	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min